FURONG JIA

jiafurong2002@gmail.com | Homepage | Google Scholar
Durham, North Carolina, United States

Research Interests

I study **how data shapes LLM behavior** and **how LLMs interpret data**, toward AI that is reliable, interpretable, and robust. I examine how training data shapes model knowledge and reasoning, how to evaluate models faithfully, and how they behave under **distribution shift** across clinical text, time series, and multimodal data. Building on this, I now study **LLM agents** for **healthcare** and **scientific discovery**: when agent design actually matters, and how agents can analyze and explain clinical data.

Education

Duke University — *Durham, NC*
Ph.D. in Computer Science — Aug. 2024 – 2029 (expected)
Advisor: Monica Agrawal — GPA: 4.0/4.0
M.S. in Computer Science (en route) — Sep. 2026

University of Southern California — *Los Angeles, CA*
B.S. in Computer Science; B.S. in Applied and Computational Mathematics — May 2024
W.V.T. Rusch Engineering Honors Program; Viterbi Grand Challenges Scholar — GPA: 3.96/4.0

Research Experience

IBM, Core AI Team — *Boston, MA*
Research Intern; Mentor: Akash Srivastava — May 2026 – Aug. 2026

- **Harnesses for LLM-driven discovery.** To separate a harness's effect from task, model, and budget, we led a controlled comparison of 11 search harnesses on mathematical construction, GPU kernel optimization, and language-model training. Harness choice often mattered more than a model upgrade, but the winner depended on the setup: counting budget in candidates instead of API cost changed the top harness in half of the settings, switching the model reversed 41% of harness pairs, and running the autoresearch loop via a coding agent, not a fixed pipeline, benefited from using the stronger model but rarely the weaker. [12]

DukeNLP, Duke University — *Durham, NC*
Advisor: Monica Agrawal — Aug. 2024 – Present

- **Probabilistic knowledge from pretraining.** We studied how faithfully LLMs acquire probabilistic knowledge from pretraining text. We built a controlled continual-pretraining framework where known data-generating networks define ground-truth distributions, synthetic clinical-style corpora verbalize sampled configurations, and OLMo-2 models are evaluated by comparing native next-token probabilities with ground-truth posteriors. We found that models recover individual probabilities well but compress the relations between them, understating how far apart outcomes are and how much evidence should move a prediction. We call this *relational compression*, which is typically under-measured by standard metrics because they score each probability in isolation. [11]
- **Interpreting dataset shift in clinical notes.** We addressed the need to explain, more than just detect, distribution shifts in clinical text. We established an extensible benchmark suite that induces synthetic shifts using real clinical notes, developed methods for evaluating generated shift explanations, and introduced *SIReNs*, an end-to-end approach that explains distributional differences by selecting representative notes from each distribution. We are also extending this work to benchmark agent capabilities and agent harnesses for interpreting shifts in multi-factorial settings and more general textual domains. [3]
- **Where LLMs learn clinical knowledge.** We examined how evidence in pretraining corpora shapes clinical LLM behavior. We linked keyword-pair occurrences in pretraining data to model performance on clinical jargon understanding and unsupported medical claims, curated *MedLingo* from real-world clinical-note abbreviations, and analyzed sources of clinical information in pretraining datasets to guide future corpus design. [4]

Melady Lab, University of Southern California — *Los Angeles, CA*
Advisor: Yan Liu — May 2023 – Jun. 2024

- **Large Language Models for Multimodal Time Series Forecasting**
  We introduced *GPT4MTS*, a prompt-based framework that fuses numeric time series with text. We also built a GDELT-derived multimodal news impact dataset and showed consistent forecasting gains over strong unimodal baselines, demonstrating effective multimodal fusion and the value of extra-textual information. [5]
  We developed *TEMPO*, a GPT architecture-based time series framework that combines seasonal trend decomposition with adaptive, selection-based prompting for non-stationary distribution shift. It achieves zero-shot state-of-the-art across diverse and unseen (including multi-modal) benchmarks, indicating strong potential as a foundation model for time series forecasting. [6]

GLAMOR Lab, University of Southern California *Los Angeles, CA*
Advisor: Jesse Thomason Aug. 2022 – Jun. 2024

- **Parameter-efficient finetuning for Multimodal Continual Learning**
  We applied parameter-efficient knowledge transfer using adapters in multimodal models for visual question answering tasks. We implemented and trained model architectures on multiple vision-language tasks, including GQA, COCO-QA, and AQUA, to enhance forward transfer and reduce knowledge forgetting. [7]

Publications

Conference Publications *Equal contribution.

[1] **Batch-of-Thought: Cross-Instance Learning for Enhanced LLM Reasoning**
Xuan Yang, **Furong Jia**, Roy Xie, Xiong Xi, Hengwei Bian, Jian Li, Monica Agrawal.
EMNLP 2026 Main

[2] **Counting Clues: A Lightweight Probabilistic Baseline Can Match an LLM**
Furong Jia*, Yuan Pu*, Finn Guo, Monica Agrawal.
AACL-IJCNLP 2026 Main

[3] **Interpreting Dataset Shift in Clinical Notes**
Shariar Vaez-Ghaemi*, **Furong Jia***, Monica Agrawal.
ML4H 2025. PMLR 297:243–262, 2026.

[4] **How does my language model understand clinical text?**
Furong Jia, David Sontag, Monica Agrawal.
CHIL 2025. PMLR 287:720–743.

[5] **GPT4MTS: Prompt-Based Large Language Model for Multimodal Time-Series Forecasting**
Furong Jia, Kevin Wang, Yixiang Zheng, Defu Cao, Yan Liu.
EAAI 2024. Proc. AAAI 38(21):23343–23351.

[6] **TEMPO: Prompt-based Generative Pre-trained Transformer for Time Series Forecasting**
Defu Cao, **Furong Jia**, Sercan Ö. Arık, Tomas Pfister, Yixiang Zheng, Wen Ye, Yan Liu.
ICLR 2024.

[7] **I2I: Initializing Adapters with Improvised Knowledge**
Tejas Srinivasan, **Furong Jia**, Mohammad Rostami, Jesse Thomason.
CoLLAs 2023. PMLR 232:923–935.

Workshop / Findings & Abstracts

[8] **What Patients Really Ask: Exploring the Effect of False Assumptions in Patient Information Seeking**
Raymond M. Xiong, **Furong Jia**, Lionel Wong, Monica Agrawal.
ML4H 2025 Findings. arXiv:2601.15674, 2026.

[9] **Leveraging Large Language Models to Identify Barriers to Kidney Transplant in the STEPS Study**
Monica Agrawal, Metin Gurcan, Sarah Peskoe, Freya Gulamali, **Furong Jia**, Tara Strigo, Ebony Boulware, Clarissa Diamantidis.
Am. J. Kidney Dis. 87(5):S203–S204, 2026. (Abstract)

Preprints

[10] **Challenges of Auditing: Variability in Outputs of Large Language Models for Health**
Yuan Pu, Yewon Chang, **Furong Jia**, Xunjian Yin, Jessica Ma, Ayman Ali, Monica Agrawal.
arXiv:2609.16590, 2026.

Manuscripts Under Review

[11] **How Faithfully do Language Models Acquire Probabilistic Knowledge?**
Furong Jia, Xunjian Yin, Yuan Pu, Monica Agrawal.

[12] **When Does the Harness Matter for LLM-Driven Discovery?**
Furong Jia, Xinheng Ding, Chengrui Qu, Monica Agrawal, Akash Srivastava.

Honors and Awards

NeurIPS: Top Reviewer (2026).

University of Southern California: Albert Dorman Future Leader Award (2024); Student Recognition Awards (2024); Provost's Research Fellowship (Fall 2023); CURVE Research Fellowship (2022–2023); Academic Achievement Award (2021); ABC Innovation Prize (2021).

Service

- **Reviewer:** NeurIPS (2025, 2026); ICLR (2027); ML4H (2025, 2026); ACL ARR: Feb./Jun./Oct. 2024, Feb./May 2025; NeurIPS FMTS Workshop (2026); COLM DAIH Workshop (2026).
- **Teaching Assistant, Duke:** Natural Language Processing (Spring 2026); Applied Machine Learning (Spring 2025).
- **Teaching Assistant, USC:** Algorithms and Theory of Computing (Spring 2022); Web Publishing and Front-end Development (Fall 2021 – Spring 2024).